Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

AI Solution Engineer with 2+ years building production GenAI applications and reusable platform capabilities using **Python**, **LLMs**, **PyTorch**, RAG, cloud services, and developer-focused documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Translated application requirements into **reusable LLM orchestration patterns** with FastAPI workers, explicit job states, failure isolation, and monitoring, processing 1K+ AI jobs at 3x throughput under concurrent load.
- Built governed data integrations across Azure SQL, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with **validation, reconciliation, CI/CD, and 99%+ production reliability** across regulated application environments.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **common GenAI model interface** across GPT, Claude, and Qwen with multimodal routing, fallbacks, timeout handling, and usage tracking, serving 10K+ daily requests for 150+ instructors.
- Developed React and Flask application patterns for uploads, rubric generation, grading, and feedback, reducing turnaround 60% while documenting workflows and troubleshooting **model, OCR, and API reliability issues**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** with evaluation pipelines over 100K+ trajectory logs, improving long-horizon planning 18% while documenting model behavior, experiments, and measurable results for autonomous research workflows.
- Compared three contemporary model architectures through controlled simulations, multi-head attention experiments, and reproducible pipelines, establishing **evaluation criteria and failure analysis** that improved closed-loop stability 15% across autonomous research workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Converted stakeholder requirements into object-oriented microservices and large-scale API integrations, improving response times 30% while collaborating with Microsoft and GitHub teams through **technical consultation, reviews, and Agile delivery**.
- Created **reusable testing and deployment patterns** with 20+ PyTest and Mockito suites, Jenkins pipelines, VM builds, and cross-platform validation, increasing pipeline efficiency 68% and reducing technical debt 20%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client requirements into **technical specifications and reusable architecture** for 12+ integrations, mapping booking, payment, coupon, and real-time status workflows before implementation to reduce design errors 25%.
- Built React and Flask application flows with Firebase services, REST APIs, WebSockets, payments, and cross-platform delivery, supporting 50K+ records with **sub-200-millisecond API responses** during client demonstrations and reviews.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL
Generative AI: GPT, LLaMA-2, Claude, Qwen, Hugging Face Transformers, RAG, Prompt Engineering, Embeddings, Multimodal AI, Model Evaluation, PyTorch, FAISS, Pinecone
Backend & Data: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, SQL Server, MySQL, Data Modeling, ETL
Cloud & DevOps: Azure, AWS, Docker, CI/CD, Jenkins, GitHub Actions, GitHub, Monitoring, Logging
Solution Engineering: Proof of Concepts, Reusable Patterns, Requirements Analysis, Technical Specifications, Developer Documentation, Troubleshooting, Unit Testing, Integration Testing, Agile, Jira, Data Governance, Data Privacy, Security, Stakeholder Management

Projects

GenAI Pattern Hub | *Python, FastAPI, GPT, LLaMA, RAG, PostgreSQL, Azure, Docker*

- Designed a **reusable GenAI solution library** with authenticated FastAPI examples for retrieval, prompting, model routing, tool integration, and human review, accompanied by decision guides and adoption-focused developer documentation.

ModelScope Lab | *Python, PyTorch, Hugging Face, FastAPI, PostgreSQL, GitHub Actions*

- Built a **model capability prototyping workbench** for comparing pretrained and fine-tuned model behavior, capturing evaluation results, inspecting failure cases, and exporting reproducible sample code for application development teams.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **GenAI retrieval application** with LLaMA-2, transformer embeddings, and FAISS semantic search over 50K+ documents with Clerk authentication, cutting retrieval latency 40% at 95% query accuracy.